THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — MATHEMATICS CHAPTER TEST SET A

Chapter 6: **Triangles** | RBSE Board Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat figures wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. All circles are: 1

(a) congruent **(b)** similar

(c) neither similar nor congruent **(d)** equal in area

2. In $\triangle ABC$, $DE \parallel BC$ with $AD = 2$ cm, $DB = 3$ cm and $AE = 4$ cm. Then EC is: 1

(a) 6 cm **(b)** 4 cm **(c)** 5 cm **(d)** 8 cm

3. If $\triangle ABC \sim \triangle PQR$ with $\angle A = 40^\circ$ and $\angle B = 60^\circ$, then $\angle R$ is: 1

(a) 40° **(b)** 60° **(c)** 80° **(d)** 100°

4. A student states: "Any two rhombuses are similar." The correct response is: 1

(a) Correct

(b) Incorrect — sides are proportional but angles may differ

(c) Correct only for squares

(d) Cannot be determined

5. If $\angle A = \angle P$ and $\angle B = \angle Q$ in $\triangle ABC$ and $\triangle PQR$, the triangles are similar by: 1

(a) SSS **(b)** SAS **(c)** AA **(d)** RHS

6. $\triangle ABC \sim \triangle DEF$ with $AB = 3$ cm, $DE = 6$ cm and $BC = 4$ cm. Then EF is: 1

(a) 2 cm **(b)** 6 cm **(c)** 8 cm **(d)** 12 cm

7. $\triangle ABC$ has sides 6, 8, 10 cm and $\triangle PQR$ has sides 3, 4, 5 cm. Student A says they are similar by SSS; Student B says they are not similar since the sides are unequal. Then: 1

(a) Student A is correct **(b)** Student B is correct

(c) Both are correct **(d)** Both are wrong

8. A vertical stick 12 m long casts a shadow 8 m long. At the same time a tower casts a shadow 40 m long. The height of the tower is: 1

(a) 60 m **(b)** 50 m **(c)** 45 m **(d)** 80 m

9. In $\triangle ABC$, $DE \parallel BC$ and $\frac{AD}{DB} = \frac{3}{5}$. Then $\frac{AE}{AC}$ is: 1

(a) $\frac{3}{5}$ **(b)** $\frac{5}{3}$ **(c)** $\frac{3}{8}$ **(d)** $\frac{5}{8}$

10. "A line through the mid-point of one side of a triangle, parallel to another side, bisects the third side." This follows from: 1

(a) Pythagoras theorem **(b)** Basic Proportionality Theorem

(c) SSS criterion **(d)** RHS criterion

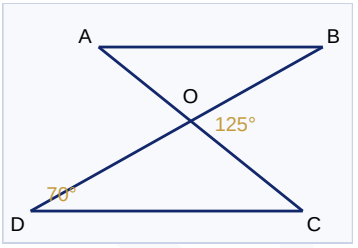
SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. E and F are points on sides PQ and PR of $\triangle PQR$. If $PE = 4$ cm, $QE = 4.5$ cm, $PF = 8$ cm and $RF = 9$ cm, state whether $EF \parallel QR$. Justify. 2

12. Give two different examples each of 2
(i) similar figures
(ii) figures that are not similar.

13. In the figure, $\triangle ODC \sim \triangle OBA$, $\angle BOC = 125°$ and $\angle CDO = 70°$. Find $\angle DOC$, $\angle DCO$ and $\angle OAB$. 2



14. A student claims: "If the corresponding sides of two triangles are proportional, the triangles are congruent." Justify or refute. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. In $\triangle ABC$, $DE \parallel BC$. If $AD = x$, $DB = x - 2$, $AE = x + 2$ and $EC = x - 1$, find the value of x. 3

16. ABCD is a trapezium in which $AB \parallel DC$ and its diagonals intersect each other at O. Show that $\frac{AO}{BO} = \frac{CO}{DO}$. 3

17. S and T are points on sides PR and QR of $\triangle PQR$ such that $\angle P = \angle RTS$. Show that $\triangle RPQ \sim \triangle RTS$. 3

OR

In $\triangle ABC$, D and E are points on AB and AC such that $AD = AE$ and $AB = AC$. Show that $\triangle ADE \sim \triangle ABC$. 3

18. A 1.5 m tall boy stands 2.4 m away from a lamp-post and his shadow is 1.6 m long. Find the height of the lamp-post. 3

OR

In $\triangle ABC$, D and E lie on AB and AC with $DE \parallel BC$. If $AD = 4$ cm, $AB = 12$ cm, $BC = 15$ cm and $AC = 9$ cm, find DE and AE. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Prove that if a line is drawn parallel to one side of a triangle to intersect the other two sides in distinct points, the other two sides are divided in the same ratio. 5

OR

ABC and AMP are two right triangles, right-angled at B and M respectively. Prove that 5
(i) $\triangle ABC \sim \triangle AMP$
(ii) $\frac{CA}{PA} = \frac{BC}{MP}$.
(iii) If $AB = 6$ cm, $AM = 4$ cm and $BC = 9$ cm, find MP.

20. **Case study:** To find the height of a tree, a boy places a small mirror on the level ground 6 m from the base of the tree. Standing 1.5 m from the mirror, he sees the top of the tree in the mirror. His eyes are 1.2 m above the ground. 5
(i) Name the similarity criterion used (angle of incidence = angle of reflection).
(ii) Find the height of the tree.
(iii) If the tree were 6.4 m tall, how far from the mirror should he stand?

OR

Sides AB and BC and median AD of $\triangle ABC$ are respectively proportional to sides PQ and QR and median PM of $\triangle PQR$. Show that $\triangle ABC \sim \triangle PQR$. 5

END OF QUESTION PAPER — **Total: 40 Marks**